PROVISIONAL PATENT APPLICATION DRAFT — FOR FILING

"CONSTANT-FOOTPRINT CORPUS COMMITMENT VIA MERKLE MOUNTAIN RANGE COVENANTS AND APPEND-ONLY MOUNTAIN UPDATES ACROSS HETEROGENEOUS DISTRIBUTED LEDGERS"

Docket: SQUIRL-OS-IP-013 (draft for filing by inventor; application serial assigned by USPTO)

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Draft date: 2026-10-20. Reduction to practice evidenced below. Prototype disclaimer applies.

This draft is not legal advice.

CROSS-REFERENCE TO RELATED APPLICATIONS

Related co-pending portfolio (same inventor): 64/157,915; 64/119,191; 64/114,746; 64/145,825;

64/174,933; 19/693,343; 64/081,490; 64/081,911; 64/082,606 (cross-ledger anchoring), and the

companion drafts SQUIRL-IP-011 ("Caging the Large Language Model") and SQUIRL-OS-IP-012

("Multi-Rail Non-Custodial Evidence Anchoring and Route Certification"). This application

is directed to the whole-corpus commitment layer practiced since 2026-10-19 (MCR-1 forward),

and incorporates the multi-rail anchoring machinery of IP-012 by reference.

FIELD

Cryptographic commitment structures; Merkle trees and Merkle Mountain Ranges; distributed-ledger

timestamping; archival integrity verification; multi-chain transaction construction.

BACKGROUND

Blockchain timestamping is sold as "permanent," but on-chain storage is neither free nor unbounded. Every byte stored on a major chain competes for scarce block space; storage cost scales linearly (or worse) with the volume of evidence a practitioner wishes to prove. Prior art therefore anchors individual documents, one digest per transaction, and faces a structural dilemma: (a) anchor little (lose provability of the corpus as a whole), or (b) anchor everything (costs grow without bound as the corpus grows).

No prior art known to the inventor commits an arbitrarily large, growing document corpus to multiple heterogeneous distributed ledgers under a fixed on-chain footprint that does not grow with the corpus, while retaining per-artifact public verifiability, and while preserving the historical integrity of every prior commitment as the corpus grows.

SUMMARY

A system and method in which the complete state of a document corpus — every file, receipt, ledger, publication, and proof bundle — is folded, at a frozen configuration, into a single Merkle Mountain Range (MMR) whose 32-byte root is the corpus's entire on-chain representation. The root is anchored, with a protocol envelope, across a certified set of heterogeneous distributed ledgers ("rails"). The on-chain footprint is constant: 32 bytes

of commitment regardless of whether the corpus holds one thousand files or one billion.

Any individual artifact remains publicly verifiable from the bare corpus plus one on-chain

root, using a logarithmic-length inclusion path (~ceil(log2 N)) — no trusted server, no proprietary index, no per-artifact chain transaction.

The mountain is append-only. When the corpus grows, the updated corpus is re-folded into a

new MMR and the new root is re-anchored across the rails ("mountain update"); every prior

anchored root remains sealed on-chain, so the update history is itself publicly verifiable,

and each update is a constant-cost event. Storage of the corpus itself is off-chain and unbounded; the chain stores only the integrity commitment. The corpus may thus be of

arbitrary size — in principle up to the size of an archive of the public internet — while each covenant event costs only the fixed envelope payload (37 bytes) plus per-rail transaction fees, on the order of satoshis.

Core doctrines reduced to practice:

- WHOLE-CORPUS COVENANT: one MMR over the complete tracked state (git tree) of the

evidence repository at a frozen commit; no cherry-picking, no partial snapshots.

- FULLY SPECIFIED MMR: leaf, node, bagging, and root constructions are defined

byte-exactly (domain-separated prefixes 0x00 / 0x01 / 0x03) so that any party can rebuild the root from the public corpus without reference implementation ambiguity.

- CONSTANT FOOTPRINT: the anchored payload is the 32-byte root in a fixed 37-byte

protocol envelope; the on-chain footprint never grows with the corpus.

- APPEND-ONLY MOUNTAIN UPDATES: growth is discharged as a new covenant (new commit, new

root, new anchor set); prior roots stay sealed on-chain; the chain of roots is a

verifiable growth history. No prior commitment is ever rewritten.

- MULTI-RAIL REDUNDANCY: the same root is anchored across rails of radically different

construction (UTXO, account-sequence, metadata-slot, remark-based, manage-data,

auxiliary-metadata) under the payload doctrine of the parent application, so no single

chain's failure, reorganization, or economic change can orphan the corpus's proof of

integrity.

- KEYLESS VERIFICATION: every anchor is read back through public keyless endpoints and

required to decode byte-exactly to the frozen root before the covenant is sealed.

- INCLUSION-PROOF RECEIPTS: designated leaves (first, last, mid, and named artifacts of

record) are proven against the root in-run and published with the runlog.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 — The covenant cycle: freeze corpus commit -> fold MMR (leaves, peaks, bag, root)

-> anchor root across N rails -> keyless read-back -> append-only seal.

FIG. 2 — The mountain-update cycle: corpus grows -> re-fold -> new root -> re-anchor ->

both roots remain sealed on-chain -> verification accepts any root in the chain.

FIG. 3 — An inclusion proof: file -> leaf hash -> O(log N) sibling path -> peak -> bag ->

root -> the 37-byte envelope on each rail.

DETAILED DESCRIPTION

The corpus and its freeze

The corpus is the set of tracked files of the evidence repository at a specific commit, enumerated in byte-sorted path order. Freezing by commit hash makes the snapshot rebuildable by any party with a copy of the repository; no private state participates.

The range construction (byte-exact)

For a corpus of N files with contents f_0 ... f_{N-1} in byte-sorted path order:

- leaf_i = SHA-256(0x00 || SHA-256(f_i)) — a domain-separated leaf over each file's

exact bytes.

- Append-only mountain: leaves are added left to right; whenever two subtrees of equal

height h exist, parent = SHA-256(0x01 || left || right), left being the earlier subtree. Peaks are the unmerged subtree roots, heights strictly decreasing left to right.

- Bagging: bag = rightmost peak; scanning right-to-left over the remaining peaks,

bag = SHA-256(0x01 || peak || bag).

- ROOT = SHA-256(0x03 || bag || N as 64-bit little-endian).

The domain-separation bytes (0x00, 0x01, 0x03) and the u64-LE leaf count bind the construction against second-preimage and length-extension ambiguity; the specification

is complete enough to re-derive the root from the public corpus alone.

The envelope and the rails

The root travels in a fixed envelope: protocol marker ("SQDR"), version byte (0x05 = root covenant; 0x06 = mountain-update covenant), and the 32-byte root — 37 bytes total,

encoded per each rail's payload doctrine (hex-ASCII where the rail requires UTF-8 memo fields; raw bytes where binary note fields are supported), as specified in the parent application. Each certified rail carries the identical envelope; the anchor set is the covenant.

Read-back and seal

Each anchor is read through a public keyless endpoint (explorer or public RPC) and must decode to the exact 37-byte envelope and the frozen root; any mismatch fails the gate.

On pass, identifiers, fees, latencies, and any failed attempts are sealed forward-only in the public ledger, and the runlog is committed with the leaf index.

The mountain update (the growth doctrine)

When the corpus grows, the method does not modify prior anchors. It re-folds the new state into a new MMR, producing a new root; the new root is anchored across the rails with the update version byte; the ledger binds the update to its parent covenant.

Verification of historical artifacts continues against the historical root; verification of new artifacts proceeds against the new root. The result: unbounded corpus growth at constant marginal proof cost, with a self-verifying history.

Scope and honesty limits

This draft claims the commitment protocol and its multi-rail discharge, not a claim that blockchains store the corpus. The data lives off-chain; the chain stores the constant-size integrity commitment. "The entire internet fits in one root" is a statement about the commitment, not on-chain storage: any off-chain archive, however large, has a 32-byte fingerprint that can be anchored for the cost of one transaction.

CLAIMS (draft)

1. A method for committing a corpus of digital artifacts to a plurality of heterogeneous

distributed ledgers under a constant on-chain footprint, comprising: folding the exact tracked state of said corpus at a frozen commit into a Merkle Mountain Range whose construction is fully specified by domain-separated hash rules; deriving a single 32-byte root; constructing a fixed protocol envelope containing said root; anchoring said envelope across a certified plurality of ledgers of differing transaction models; and verifying each anchor through keyless public reads requiring byte-exact decode.

2. The method of claim 1, wherein said Merkle Mountain Range construction comprises:

leaf hashes of the form SHA-256(0x00 || SHA-256(file bytes)) over files in byte-sorted path order; internal nodes of the form SHA-256(0x01 || left || right) for equal-height adjacent subtrees, left being the earlier subtree; a bagging of peaks right-to-left under the same node rule; and a root of the form SHA-256(0x03 || bag || leaf count as 64-bit little-endian).

3. The method of claim 1, wherein growth of said corpus is discharged by re-folding the

updated state into a new root and anchoring said new root while all prior anchored roots remain unmodified on-chain, producing an append-only, publicly verifiable history of corpus states.

4. The method of claim 1, wherein the on-chain representation of said corpus is exactly

one 32-byte value per rail regardless of the size or growth of said corpus.

5. The method of claim 1, wherein any single artifact of said corpus is verifiable from

the public corpus, any anchored root, and a logarithmic-length inclusion path, without

per-artifact ledger transactions and without trusted third parties.

6. The method of claim 1, wherein said rails include at least two of: a UTXO model rail

carrying the envelope in an OP_RETURN-style output; an account-sequence rail carrying

the envelope in transaction calldata; a rail carrying the envelope in a memo field;

a rail carrying the envelope in a remark; a rail carrying the envelope in managed

account data; and a rail carrying the envelope in auxiliary transaction metadata.

7. The method of claim 1, wherein a distinguished set of leaves, including at least the

first leaf, the last leaf, a middle leaf, and a designated artifact of record, are

proven against said root and published as inclusion-proof receipts contemporaneous

with the covenant.

8. A system comprising: a corpus under version control; a deterministic MMR folder

implementing the construction of claim 2; a plurality of certified rail writers

holding only sufficient balance for transaction fees; and a verification module

performing keyless byte-exact read-back, such that no custodian of the corpus can

alter any artifact without changing the root and thereby falsifying every rail

simultaneously.

9. The method of claim 3, wherein each update covenant is bound to its parent covenant

by a public ledger entry referencing both roots, such that the chain of roots is

itself a verifiable object.

10. The method of claim 1, wherein the protocol envelope comprises a version byte

distinguishing a first anchoring of a root from a mountain-update anchoring of a

subsequent root.

ABSTRACT

A whole-corpus cryptographic commitment system in which a document repository's complete

frozen state is folded into a Merkle Mountain Range and represented on heterogeneous

distributed ledgers by a single 32-byte root in a fixed 37-byte envelope, giving

constant on-chain footprint independent of corpus size; corpus growth is discharged by

append-only mountain updates whose prior roots remain sealed on-chain; and any

individual artifact remains publicly verifiable from the corpus, one anchored root, and

a logarithmic inclusion path.

REDUCTION-TO-PRACTICE EVIDENCE CHAIN (receipts-first, all public)

- MCR-1 "Merkle Root Covenant" (2026-10-19): 1,891-leaf MMR over the four-horsemen

evidence repository at commit 7c492528, root

2d1a32145cf89a6b7909d3059abe560e6f3d908e7d510525f4457e096f2cc23a anchored across

all ten certified mainnet rails (BTC, XRP, ETH, SOL, ADA, ALGO, AVAX, DOT, DOGE, XLM),

byte-exact keyless read-back on each; public ledger entry 0260; the present filing's

MMR construction is the MCR-1 construction, re-validated byte-exact against the frozen

root before the update ceremony of 2026-10-20.

- MCR-2 "Mountain Update" (2026-10-20, this filing's DOI ceremony): corpus grown to

~1,905 files including this patent record; new root anchored across the same ten

rails; prior MCR-1 root unmodified on every rail; update bound in the public ledger.

- Payload doctrine, thesis v2.1 DOI 10.5281/zenodo.23482978 (Section 2.5, 37-byte

envelope conventions); routing chapter DOI 10.5281/zenodo.23485222; DCAI corpus community (Zenodo).

- Route certification: DeWR-1 (10-rail certification), CFR-9 (10-leg crossfire, 10/10 confirmed), SBR-1 (XRP rail-of-record / Stellar transport burst, 8 anchors, ~4 seconds, ~one-hundredth of a cent) — all sealed in the public ledger.

FILING NOTES (inventor steps; verify against the current USPTO fee schedule)

1. Patent Center -> New application -> Provisional (SB/16).
2. Title as above; inventor Leon Calvin Long, II; small entity; pro se.
3. Upload this specification PDF as the only attachment; use the web cover sheet.
4. Copy the serial number; send to the agent chat for the IP ledger update.

DISCLOSURE DOCTRINE

AI-assisted, inventor-directed drafting under the receipts-first program: every reduction-to-practice claim above is anchored to a public, keyless-verifiable transaction. Errors, if found, are disclosed forward-only in the public ledger; the append-only doctrine applies to this draft. Dual Mesh Benchmark: NOT MEASURED (out of

scope for this filing). Prototype disclaimer: the referenced software is a research prototype, not a production system.